Downloaded from rspa.royalsocietypublishing.org

PROCEEDINGS OF THE ROYAL SOCIETY A MATHEMATICAL, PHYSICAL & ENGINEERING SCIENCES

# Transport Properties of Regular Arrays of Cylinders

W. T. Perrins, D. R. McKenzie and R. C. McPhedran

*Proc. R. Soc. Lond. A* 1979 **369**, 207-225
doi: 10.1098/rspa.1979.0160

**References**

Article cited in:
**http://rspa.royalsocietypublishing.org/content/369/1737/207#related-urls**

**Email alerting service**

Receive free email alerts when new articles cite this article - sign up in the box at the top right-hand corner of the article or click **here**

To subscribe to *Proc. R. Soc. Lond. A* go to: **http://rspa.royalsocietypublishing.org/subscriptions**

This journal is © 1979 The Royal Society

Downloaded from rspa.royalsocietypublishing.org

*Proc. R. Soc. Lond.* A **369**, 207–225 (1979)
*Printed in Great Britain*

# Transport properties of regular arrays of cylinders

By W. T. Perrins, D. R. McKenzie and R. C. McPhedran
*School of Physics, University of Sydney, New South Wales* 2006, *Australia*

(*Communicated by R. H. Brown, F.R.S. – Received* 20 *February* 1979 – *Revised* 22 *May* 1979)

We extend a method devised by Lord Rayleigh to enable the calculation of the transport properties of circular cylinders in square and hexagonal arrays. The theory is confirmed by measurements on arrays of perfectly conducting cylinders, and also is compared with asymptotic formulae due to Keller (1963) and O'Brien (1977). It is used to furnish plots of equipotential lines within the array. It is also applied to the calculation of the optical properties of films with columnar structure. Detailed studies for copper films show both the good solar selectivity possible with voided structures and the transition from a good reflector to a metal black consequent upon structural changes.

## 1. Introduction

The transport properties of circular cylinders packed in regular arrays are of considerable interest in a number of fields. The transport property may be the electrical or thermal conductivity, the dielectric permittivity, the Lamé constants or any other of the eight analogous quantities listed by Batchelor (1974). The result is of interest in thin film physics, where films consisting of columns of one material in a matrix of another are observed. It is often desirable to calculate their optical properties (see, for example, Netterfield 1976). In the field of materials physics two phase materials containing rod- or fibre-like inclusions often occur. Knowledge of their electrical or thermal conductivities and their elastic properties is valuable (Desorbo & Kline, for example, discuss rod alloys). The tendency to order in a two-dimensional system of disks when packed under gravity is much stronger than in the corresponding three-dimensional system with spheres (Visscher & Bosterli 1972). Calculations on ordered arrays of cylinders are therefore directly relevant to practical situations.

The problem of calculating the transport properties will be discussed here in the context of electrical conductivity. However, the mathematics we use and the results we obtain are immediately applicable to the seven other associated situations.

There are five possible ways of packing cylinders in regular arrays in two dimensions (*International tables for X-ray crystallography* 1962). Here we concentrate



The Royal Society is collaborating with JSTOR to digitize, preserve, and extend access to *Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences.* www.jstor.org

Downloaded from rspa.royalsocietypublishing.org



on the two lattices that are isotropic for fields applied in the plane perpendicular to the cylinder axes. We have already presented elsewhere preliminary results for the square array (Perrins *et al.* 1979), together with an evaluation of the effects of cylindrical, perfectly insulating voids in a matrix of unit conductivity. Here we will discuss briefly the theoretical methods for this array and we will consider in detail the hexagonal array. The theoretical calculations are verified by comparison with experimental measurements. A reciprocal relation due to Keller (1963, 1964) is generalized for the case of complex conductivities and extended to the hexagonal array.

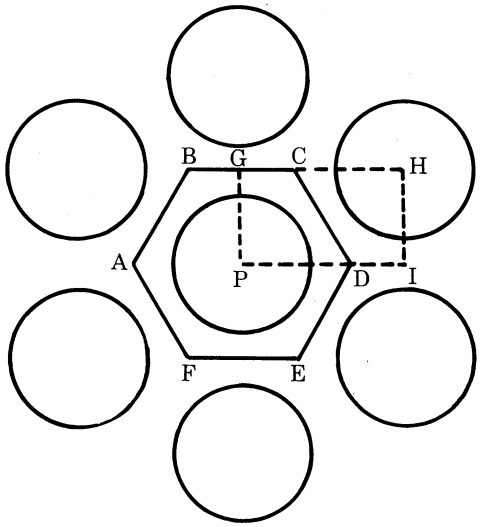



FIGURE 1. A portion of the hexagonal array of cylinders is shown, with the hexagonal unit cell ABCDEF in heavy outline and the sub-unit GHIP in dashed outline.

The case of cylinders very close to touching is amenable to asymptotic analysis (O'Brien 1977). Our results are shown to be in agreement with asymptotic formulae in this limit. As an application of the work we calculate the optical reflectivity of a film consisting of copper cylinders surrounded by vacuum (the cylinder axes being normal to the film surface).

## 2. THE HEXAGONAL ARRAY

Consider a hexagonal lattice of cylinders of radius $a$, as shown in figure 1. We may assume unit distance between the centres of neighbouring cylinders without loss of generality. The conductivities of the cylinder material and of the surrounding matrix material are specified by their ratio $\sigma$. We wish to calculate the ratio $\epsilon$ of the composite material conductivity to that of the matrix. In general $\sigma$ and $\epsilon$ are complex.

The method we use is due to Lord Rayleigh (1892). It has also been applied to

Downloaded from rspa.royalsocietypublishing.org



arrays of cylinders by Runge (1925) and by Ninham & Sammut (1976). Rayleigh's method has been extended to include an arbitrary number of terms. This approach has been used in studies of lattices of spheres by McPhedran & McKenzie (1978), by McKenzie *et al.* (1978) and by Doyle (1978).

We apply an electric field parallel to the line PD of figure 1, this line being defined as the axis of $x$. PG is the axis of $y$. The magnitude of the applied field is $E_0$, exterior to the region occupied by cylinders. Using polar coordinates $(r, \theta)$ where $r$ is measured from a cylinder centre (P) and $\theta$ from PD, following Rayleigh (1892) we expand the potential $V(r, \theta)$ about P:

$$V_c(r, \theta) = A_0 + \sum_{l=1}^{\infty} (A_l r^l + B_l / r^l) \cos l\theta \tag{1}$$

for $r > a$, and

$$V_s(r, \theta) = C_0 + \Sigma C_l r^l \cos l\theta \tag{2}$$

for $r < a$. The coefficients $A_l$ and $C_l$ are expressible in terms of the $B_l$ by using the boundary conditions at the cylinder surface:

$$\sigma \partial V_s / \partial n = \partial V_c / \partial n \tag{3}$$

and

$$V_s = V_c. \tag{4}$$

Setting $V_s(0, \theta) = 0$, we find that $C_0$ and $A_0$ are zero. Only those terms in (1) and (2) for which $l$ is odd are allowed in the expansion, in order to preserve the antisymmetric property of the potential around the origin. In addition, all terms in which $l$ is a multiple of three are disallowed, as can be seen from the following argument using the symmetry of the array. We apply, in addition to the field $E_0$ directed along PD, fields $E_0(-\frac{1}{2}, -\frac{1}{2}\sqrt{3})$ and $E_0(-\frac{1}{2}, \frac{1}{2}\sqrt{3})$ so that the total applied field is zero. If $V_c(r, \theta)$ is the potential due to $E_0$ alone, then the symmetry of the array gives values $V_c(r, \theta + \frac{2}{3}\pi)$ and $V_c(r, \theta + \frac{4}{3}\pi)$ for the potentials at $(r, \theta)$ due to the other two fields respectively. Hence

$$V_c(r, \theta) + V_c(r, \theta + \tfrac{2}{3}\pi) + V_c(r, \theta + \tfrac{4}{3}\pi) = 0. \tag{5}$$

Substituting from (1), we find

$$\sum_l (A_l r^l + B_l / r^l) \cos l\theta [1 + 2 \cos \tfrac{2}{3}\pi l] = 0. \tag{6}$$

The term in square brackets is zero for all odd $l$ except multiples of 3. Since (6) is true for all $r, \theta$, we have the result that $A_l = B_l = 0$ for $l$ a multiple of 3.

The unknowns $B_{2m-1}$, for $m = 1, 2, \ldots$, up to any desired order $N$, are obtained by the appropriate truncation and solution of the following linear equations for $n = 1, 2, \ldots, N$:

$$E_0 \delta_{n,1} - (2n-1)! \frac{(1+\sigma)}{(1-\sigma)} \frac{B_{2n-1}}{a^{4n-2}} = \sum_{m=1,2,\ldots} \frac{(2n+2m-3)!}{(2m-2)!} S_{2n+2m-2} B_{2m-1}. \tag{7}$$

Here, $\delta_{n,1}$ is the Kronecker delta symbol, being unity for $n = 1$ and zero otherwise.

Downloaded from rspa.royalsocietypublishing.org



The volume fraction $f$ of the region occupied by cylinders is related to their radius $a$ by

$$f = \tfrac{2}{\sqrt{3}}\pi a^2. \tag{8}$$

The cylinders touch when $a = \frac{1}{2}$, the volume fraction then taking the critical value $f_c = \pi/2\sqrt{3}$.

TABLE 1. VALUES OF CONDUCTIVITY FOR THE HEXAGONAL ARRAY FOR VARIOUS VALUES OF THE CYLINDER CONDUCTIVITY ($\sigma$) AND VARIOUS VOLUME FRACTIONS ($f$)

| | $\sigma$ | | | | | | |
|---|---|---|---|---|---|---|---|
| $f$ | 2 | 3.5 | 5 | 10 | 20 | 50 | $\infty$ |
| 0.1 | 1.0690 | 1.1176 | 1.1429 | 1.1782 | 1.1990 | 1.2126 | 1.2222 |
| 0.2 | 1.1429 | 1.2500 | 1.3077 | 1.3913 | 1.4419 | 1.4757 | 1.5000 |
| 0.3 | 1.2222 | 1.4000 | 1.5000 | 1.6506 | 1.7451 | 1.8100 | 1.8572 |
| 0.4 | 1.3077 | 1.5715 | 1.7274 | 1.9733 | 2.1348 | 2.2490 | 2.3340 |
| 0.5 | 1.4001 | 1.7696 | 2.0008 | 2.3865 | 2.6551 | 2.8530 | 3.0047 |
| 0.6 | 1.5002 | 2.0016 | 2.3368 | 2.9376 | 3.3901 | 3.7433 | 4.0267 |
| 0.65 | 1.5536 | 2.1336 | 2.5363 | 3.2906 | 3.8883 | 4.3738 | 4.7760 |
| 0.7 | 1.6095 | 2.2785 | 2.7631 | 3.7190 | 4.5252 | 5.2147 | 5.8111 |
| 0.73 | 1.6443 | 2.3727 | 2.9152 | 4.0245 | 5.0031 | 5.8733 | 6.6524 |
| 0.76 | 1.6800 | 2.4731 | 3.0818 | 4.3778 | 5.5830 | 6.7076 | 7.7600 |
| 0.78 | 1.7045 | 2.5439 | 3.2022 | 4.6467 | 6.0459 | 7.4036 | 8.7223 |
| 0.80 | 1.7295 | 2.6182 | 3.3311 | 4.9489 | 6.5907 | 8.2600 | 9.9586 |
| 0.82 | 1.7550 | 2.6962 | 3.4699 | 5.2924 | 7.2452 | 9.3479 | 11.6207 |
| 0.84 | 1.7810 | 2.7785 | 3.6202 | 5.6887 | 8.0536 | 10.7931 | 14.0093 |
| 0.85 | 1.7943 | 2.8213 | 3.7002 | 5.9117 | 8.5374 | 11.7200 | 15.6656 |
| 0.86 | 1.8077 | 2.8655 | 3.7840 | 6.1554 | 9.0930 | 12.8495 | 17.8381 |
| 0.87 | 1.8212 | 2.9110 | 3.8719 | 6.4237 | 9.7427 | 14.2733 | 20.8570 |
| 0.88 | 1.8349 | 2.9579 | 3.9645 | 6.7225 | 10.5216 | 16.1565 | 25.4508 |
| 0.89 | 1.8488 | 3.0064 | 4.0623 | 7.0606 | 11.4922 | 18.8532 | 33.7007 |
| 0.895 | 1.8558 | 3.0313 | 4.1136 | 7.2486 | 12.0856 | 20.7614 | 41.3407 |
| 0.90 | 1.8628 | 3.0568 | 4.1665 | 7.454 | 12.790 | 23.397 | 56.229 |
| 0.905 | 1.8699 | 3.0826 | 4.2213 | 7.679 | 13.67 | 27.68 | 112.8 |
| 0.9068 | 1.8725 | 3.0921 | 4.2416 | 7.769 | 14.07 | 30.39 | — |
| 0.9068993 | 1.8726 | 3.0926 | 4.2427 | 7.774 | 14.09 | 30.59 | — |

In (7), the quantities $S_l$ are sums of the form

$$S_l = \sum_{j=1}^{\infty} (x_j + iy_j)^{-l}, \tag{9}$$

where as $j$ runs from 1 to $\infty$, $(x_j, y_j)$ runs over all cylinder centres, apart from the origin, P. Although the sum is written, following Rayleigh, in complex form, all non-zero $S_l$ in this problem are real. The sum $S_2$ is conditionally convergent; its value is dependent upon the shape of the exterior boundary of the array. The shape dependence may be removed from the formalism in the same way as for lattices of spheres (McPhedran & McKenzie 1978). A polarization field $E_P$ is defined, which incorporates the effects of polarization charges on the exterior boundary of the array. $E_0$ in (7) is replaced by

$$E = E_0 + E_P, \tag{10}$$

Downloaded from rspa.royalsocietypublishing.org



where $E$ is the total macroscopic field inside the array (in the neighbourhood of P). $S_2$ then always has the value $2\pi/\sqrt{3}$. This value is obtained by summation over a 'needle'-shaped region, infinitely more extended along the $x$ axis than along the $y$ axis, in which case $E_P = 0$. The other sums $S_l$ for $l > 2$ are absolutely convergent and are obtained by direct summation. Values of the sums are given in appendix 1.

By application of the Green theorem to the unit cell ABCDEF of figure 1, the conductivity, $\epsilon$, of the array can be derived:

$$\epsilon = 1 - 4\pi B_1/\sqrt{3}\,E. \tag{11}$$

TABLE 2. THE MINIMUM ORDER $N$ OF THE SOLUTION (FOR THE HEXAGONAL ARRAY) REQUIRED TO GIVE CONDUCTIVITY TO THE ACCURACY QUOTED IN TABLE 1. $N$ IS LISTED FOR VARIOUS VALUES OF $f$ AND $\sigma$

| | $\sigma$ | | | | | | |
|---|---|---|---|---|---|---|---|
| $f$ | 2 | 3.5 | 5 | 10 | 20 | 50 | $\infty$ |
| 0.1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 0.2 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 0.3 | 1 | 1 | 1 | 1 | 1 | 1 | 3 |
| 0.4 | 1 | 3 | 3 | 3 | 3 | 3 | 3 |
| 0.5 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| 0.6 | 3 | 3 | 3 | 3 | 3 | 3 | 6 |
| 0.7 | 3 | 3 | 6 | 6 | 6 | 6 | 6 |
| 0.76 | 3 | 6 | 6 | 9 | 9 | 9 | 12 |
| 0.8 | 3 | 6 | 9 | 12 | 12 | 12 | 15 |
| 0.84 | 3 | 9 | 12 | 12 | 15 | 18 | 18 |
| 0.86 | 6 | 9 | 12 | 15 | 18 | 18 | 24 |
| 0.88 | 6 | 12 | 12 | 21 | 24 | 30 | 36 |
| 0.89 | 6 | 12 | 15 | 24 | 30 | 36 | 48 |
| 0.90 | 6 | 15 | 15 | 30 | 42 | 51 | 90 |
| 0.9068993 | 6 | 15 | 24 | 39 | 54 | 99 | — |

It is an interesting fact that $\epsilon$ is a monotonically increasing function of the order of solution $N$, for real $\sigma$. Thus, our numerical values converge from below to the true solution, as $N$ increases.

Table 1 gives values of $\epsilon$ for various values of $f$ and real values of $\sigma$. The above method was used with a square truncation of equations (7). Table 2 shows the minimum order $N$required if the solution is to be accurate to the number of figures shown. This order increases with both cylinder conductivity, $\sigma$, and volume fraction, $f$.

For problems in which $\sigma$ is small or the cylinders cannot approach each other closely, a low order solution is useful. We give here the analytic formula for the solution of order $N = 4$, based upon square truncation of the equations (7):

$$\epsilon = 1 - 2f\Big/\left[T + f - \frac{0.075422 f^6 T}{T^2 - 1.060283 f^{12}} - \left\{\frac{0.000076 f^{12}}{T}\right\}\right], \tag{12}$$

where

$$T = (1+\sigma)/(1-\sigma). \tag{13}$$

Downloaded from rspa.royalsocietypublishing.org



The term in curly brackets is the first correction term obtained by a triangular truncation for order $N = 6$. The region of applicability of (12) may be determined by reference to table 2.

## 3. The square array

The theory for the square array has already been discussed (Perrins *et al.* 1979). The values of the sums $S_l$ for the square array are given in appendix 2.

Table 3. Values of conductivity for the square array for various values of the cylinder conductivity ($\sigma$) and various volume fractions ($f$)

| | $\sigma$ | | | | | | |
|---|---|---|---|---|---|---|---|
| $f$ | 2 | 3.5 | 5 | 10 | 20 | 50 | $\infty$ |
| 0.1 | 1.0690 | 1.1176 | 1.1429 | 1.1782 | 1.1990 | 1.2126 | 1.2222 |
| 0.2 | 1.1429 | 1.2500 | 1.3078 | 1.3915 | 1.4421 | 1.4760 | 1.5003 |
| 0.3 | 1.2223 | 1.4004 | 1.5007 | 1.6520 | 1.7472 | 1.8125 | 1.8602 |
| 0.4 | 1.3080 | 1.5732 | 1.7307 | 1.9806 | 2.1459 | 2.2633 | 2.3510 |
| 0.5 | 1.4010 | 1.7756 | 2.0130 | 2.4155 | 2.7011 | 2.9146 | 3.0802 |
| 0.55 | 1.4507 | 1.8912 | 2.1813 | 2.6948 | 3.0771 | 3.3732 | 3.6094 |
| 0.6 | 1.5028 | 2.0190 | 2.3744 | 3.0372 | 3.5618 | 3.9881 | 4.3418 |
| 0.64 | 1.5464 | 2.1319 | 2.5518 | 3.3764 | 4.0715 | 4.6674 | 5.1844 |
| 0.67 | 1.5803 | 2.2240 | 2.7020 | 3.6854 | 4.5659 | 5.3623 | 6.0875 |
| 0.7 | 1.6154 | 2.3239 | 2.8709 | 4.0617 | 5.2137 | 6.3359 | 7.4327 |
| 0.72 | 1.6394 | 2.3954 | 2.9964 | 4.3655 | 5.7806 | 7.2573 | 8.8063 |
| 0.74 | 1.6641 | 2.4716 | 3.1348 | 4.7303 | 6.5263 | 8.5910 | 11.0062 |
| 0.75 | 1.6767 | 2.5117 | 3.2099 | 4.9443 | 7.0041 | 9.5355 | 12.7511 |
| 0.76 | 1.6894 | 2.5533 | 3.2900 | 5.1867 | 7.5916 | 10.8187 | 15.4412 |
| 0.77 | 1.7023 | 2.5965 | 3.3747 | 5.4674 | 8.3505 | 12.7411 | 20.4334 |
| 0.78 | 1.7154 | 2.6415 | 3.4663 | 5.8037 | 9.427 | 16.310 | 35.934 |
| 0.785 | 1.7220 | 2.6648 | 3.5150 | 6.004 | 10.23 | 20.5 | — |
| 0.785 398 | 1.7226 | 2.6667 | 3.5190 | 6.022 | 10.31 | 21.1 | — |

In table 3 we list values of $\epsilon$ for various real values of $\sigma$ and various volume fractions, $f$. Table 4 gives the minimum order, $N$, of solution required to achieve the accuracy shown in table 3.

We give an analytic expression for the conductivity of this array in a similar form to Rayleigh (1892) based on square truncation to order $N = 3$:

$$\epsilon = 1 - 2f \Big/ \left[ T + f - \frac{0.305827 f^4 T}{T^2 - (1.402958 f^8)} - \left\{ \frac{0.013362 f^8}{T} \right\} \right], \tag{14}$$

where $T$ is again defined by equation (13).

The term in curly brackets is the first correction term obtained from triangular truncation for order $N = 4$. It is to be noted that Rayleigh failed to include the term in round brackets in his analytic expression based on the same order of summation. The region of applicability of (14) may be determined by reference to table 4.

Downloaded from rspa.royalsocietypublishing.org



## 4. Experimental confirmation of the theory

The experimental technique used for the hexagonal array was similar to that employed in measurements made on a body centred cubic lattice of spheres (McKenzie *et al.* 1978). A set of six perspex boxes were constructed so that their form corresponded to the subunit GHIP of the cell of the array (figure 1). Brass plates were used to give the equipotential planes PG and IH, quarter cylinders being attached to them

Table 4. Minimum values of $N$ required (for the square array) to obtain the accuracy displayed in table 3. $N$ is listed for various cylinder conductivities ($\sigma$) and various volume fractions ($f$)

| | $\sigma$ | | | | | | |
|---|---|---|---|---|---|---|---|
| $f$ | 2 | 3.5 | 5 | 10 | 20 | 50 | $\infty$ |
| 0.1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 0.2 | 1 | 1 | 2 | 2 | 2 | 2 | 2 |
| 0.3 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| 0.4 | 2 | 2 | 2 | 2 | 4 | 4 | 4 |
| 0.5 | 2 | 4 | 4 | 4 | 4 | 4 | 6 |
| 0.6 | 4 | 6 | 6 | 6 | 6 | 8 | 8 |
| 0.7 | 4 | 8 | 8 | 10 | 12 | 12 | 16 |
| 0.75 | 6 | 10 | 12 | 18 | 20 | 24 | 36 |
| 0.77 | 6 | 12 | 16 | 24 | 36 | 42 | 40 |
| 0.78 | 8 | 12 | 20 | 36 | 36 | 48 | 100 |
| 0.785 | 8 | 14 | 20 | 36 | 48 | 60 | — |
| 0.785398 | 12 | 16 | 24 | 48 | 54 | 60 | — |

Table 5. Experimental measurements of the conductivity ($\epsilon_{\text{expt}}$) of an hexagonal array of cylinders in a matrix compared with the theoretical result ($\epsilon_{\text{theor}}$) at seven volume fractions ($f$)

| $f$ | $\epsilon_{\text{expt}}$ | $\epsilon_{\text{theo}}$ |
|---|---|---|
| $0.651 \pm 0.002$ | $4.93 \pm 0.15$ | 4.80 |
| $0.754 \pm 0.002$ | $7.58 \pm 0.23$ | 7.51 |
| $0.802 \pm 0.002$ | $10.34 \pm 0.31$ | 10.08 |
| $0.849 \pm 0.002$ | $16.12 \pm 0.48$ | 15.41 |
| $0.873 \pm 0.002$ | $22.79 \pm 0.68$ | 22.10 |
| $0.881 \pm 0.002$ | $27.39 \pm 0.82$ | 26.13 |
| $0.894 \pm 0.002$ | $34.00 \pm 1.02$ | 39.11 |

at P and H. The conductivity of each box, when filled with water, was measured by using an a.c. bridge at 20 kHz, both with and without the quarter cylinders. The conductivity ratios ($\epsilon_{\text{expt}}$) represent the composite material conductivity, and these are compared with the theoretical values ($\epsilon_{\text{theor}}$) in table 5 and figure 2. The experimental measurements of conductivity are accurate to $\pm 3\,\%$ while volume fractions are correct to $\pm 0.002$.

Experimental measurements for the square array have been given elsewhere

Downloaded from rspa.royalsocietypublishing.org



(Perrins *et al.* 1979) but for completeness they are included in figure 2. Also on figure 2 are experimental results of Kharadly & Jackson (1952), who measured the permittivities of square arrays of brass and steel cylinders in a Polyfoam matrix (for volume fractions up to 0.66).

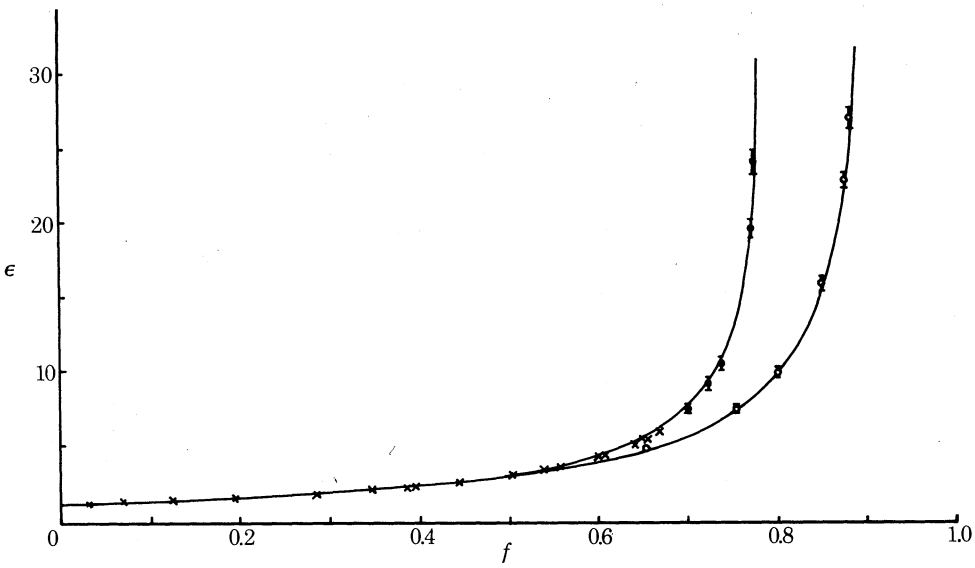



FIGURE 2. The array conductivity, $\epsilon$, is shown as a function of volume fraction, $f$, for the square array and the hexagonal array. The solid lines are the theoretical curves. Experimental measurements are indicated thus: ×, measurements of Kharadly & Jackson (1952) for the square array; ●, measurements of Perrins *et al.* (1979) for the square array; ○, new measurements for the hexagonal array.

## 5. Arrays of cylinders in contact

A limiting case of interest is that in which the cylinders are in contact. Asymptotic analysis of O'Brien (1977) shows that, for a pair of cylinders of conductivity $\sigma$ separated by a small gap $h$, the current flowing in response to unit applied field is

$$I(\sigma, h) \simeq \left(\frac{2}{\pi}\frac{\ln \sigma}{\sigma} + \frac{1}{\pi}\sqrt{\frac{h}{a}}\right)^{-1} \tag{15}$$

for $\sigma \gg 1$. For a square array of cylinders, $I$ is also the total current flowing in the unit cell, provided that the contribution to $\epsilon$ from the region around the touching line outweighs that from other regions. For an hexagonal array of cylinders, $I$ must be multiplied by $\sqrt{3}$ to obtain $\epsilon$. (This latter result is readily derived by using a simple argument in which close contacts between cylinders are represented by resistance values calculated from (15).)

Downloaded from rspa.royalsocietypublishing.org



For arrays of touching cylinders ($h = 0$) (15) gives

$$\epsilon_S \simeq \tfrac{1}{2}\pi\sigma/\ln\sigma \tag{16}$$

and

$$\epsilon_H \simeq \tfrac{1}{2}\sqrt{3}\,\pi\sigma/\ln\sigma, \tag{17}$$

where $\epsilon_S$ and $\epsilon_H$ denote the square and hexagonal array conductivities respectively. These expressions are only the first terms in series, with subsequent terms depending on details of array geometry not included in O'Brien's analysis.

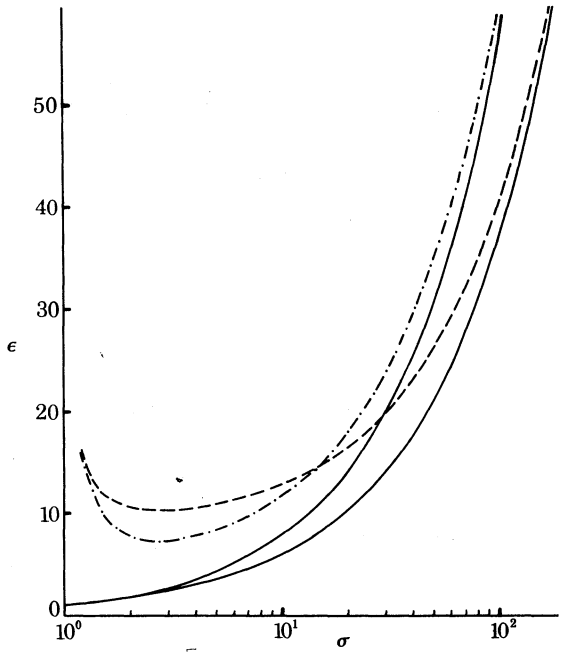



FIGURE 3. A comparison of the asymptotic formulae of O'Brien (broken curves) with the exact results for both square and hexagonal arrays (solid curves). – – –, Asymptotic formula (16) with an additive constant of 6.0 for the square array; –·–·–, asymptotic formula (17) for the hexagonal array.

In figure 3 the computed results of conductivity, for both square and hexagonal arrays of cylinders in contact, are shown as a function of cylinder conductivity. The curves corresponding to the asymptotic formulae (16) and (17) are also shown for comparison. The curve for the square array (15) has been plotted with an additive constant of 6.0. This is a numerical estimate for the term of order $\sigma^0$ in the series whose first term is (16). No additive constant has been used in the case of the asymptotic formula for the hexagonal array, the term in $\sigma^0$ evidently being small in this case.

The calculation of exact values of conductivity requires excessive amounts of computing effort for cylinder conductivities beyond about 200. However, figure 3

Downloaded from rspa.royalsocietypublishing.org



shows that the exact results and the asymptotic curves agree well in slope for values of $\sigma$ in excess of 80. Consequently (16) and (17) will agree precisely with the corresponding exact results for very large $\sigma$, when the effect of the additive constant used in plotting (16) vanishes. If we call the region in which (16) and (17) become exact to within 1 or 2 % the 'asymptotic region', we can estimate the value of $\sigma$ required to attain it by taking the ratio of the conductivities of the hexagonal and square arrays for the same gap $h$ between cylinders. This ratio has the value $\sqrt{3}$ in the asymptotic region. Observation of table 6 shows that the tendency of this ratio to $\sqrt{3}$ is clear both in the case where $h$ is zero with $\sigma$ varying, and in the case where $h$ varies with $\sigma$ infinite. However, it can be seen that in the case of $h = 0$, the asymptotic region is not attained until $\sigma$ greatly exceeds 200.

TABLE 6. THE MANNER IN WHICH $R$ APPROACHES ITS LIMITING VALUES OF $\sqrt{3}$ ($\approx 1.732$) IS SHOWN. $R$ REPRESENTS THE RATIO OF THE CONDUCTIVITY OF THE HEXAGONAL ARRAY TO THAT OF THE SQUARE ARRAY.

| $\sigma = \infty$ | | $h = 0$ | |
|---|---|---|---|
| $h$ | $R$ | $\sigma$ | $R$ |
| 0.1 | 1.34 | 10 | 1.29 |
| 0.03 | 1.50 | 20 | 1.37 |
| 0.01 | 1.60 | 50 | 1.45 |
| 0.005 | 1.639 | 60 | 1.46 |
| 0.003 | 1.660 | 70 | 1.47 |
| 0.002 | 1.674 | 100 | 1.50 |

The first table is for perfectly conducting cylinders ($\sigma = \infty$) as the gap ($h$) between cylinders approaches zero.

The second table is for the case when the gap ($h$) is zero, and the cylinders are in contact, as their conductivity $\sigma$ is increased.

For perfectly conducting cylinders, the result of O'Brien (1977) is equivalent to that of Keller (1963). We have noted elsewhere (Perrins *et al.* 1979) that Keller's formula agrees with our computed results for the square array to within 4 % at $f = 0.782$.

## 6. POTENTIAL PLOTS AND THE DEPOLARIZATION FACTOR

Equations (1) and (2) enable the distribution of potentials throughout the unit cell of either array to be found by using the appropriate values of the coefficients $B_l$ determined by the solution of the equations (7). It is instructive to plot these distributions and we have done so in figure 4 for three values of conductivity $\sigma$ and for both square and hexagonal arrays.

Consider first the case of the square array. For low volume fractions, interaction effects between neighbouring cylinders are small and the electric field inside each cylinder is practically homogeneous. The potential distribution at the surface of a

Downloaded from rspa.royalsocietypublishing.org



typical cylinder $V(a, \theta)$ follows closely a cosine dependence upon angle. For larger volume fractions, interaction effects increase, the field departs from uniformity and the cosine dependence of $V(a, \theta)$ breaks down. It is interesting to note the concentration of field lines in the vicinity of the point of nearest approach. This concentration is of course accompanied by a localization of surface charge density.

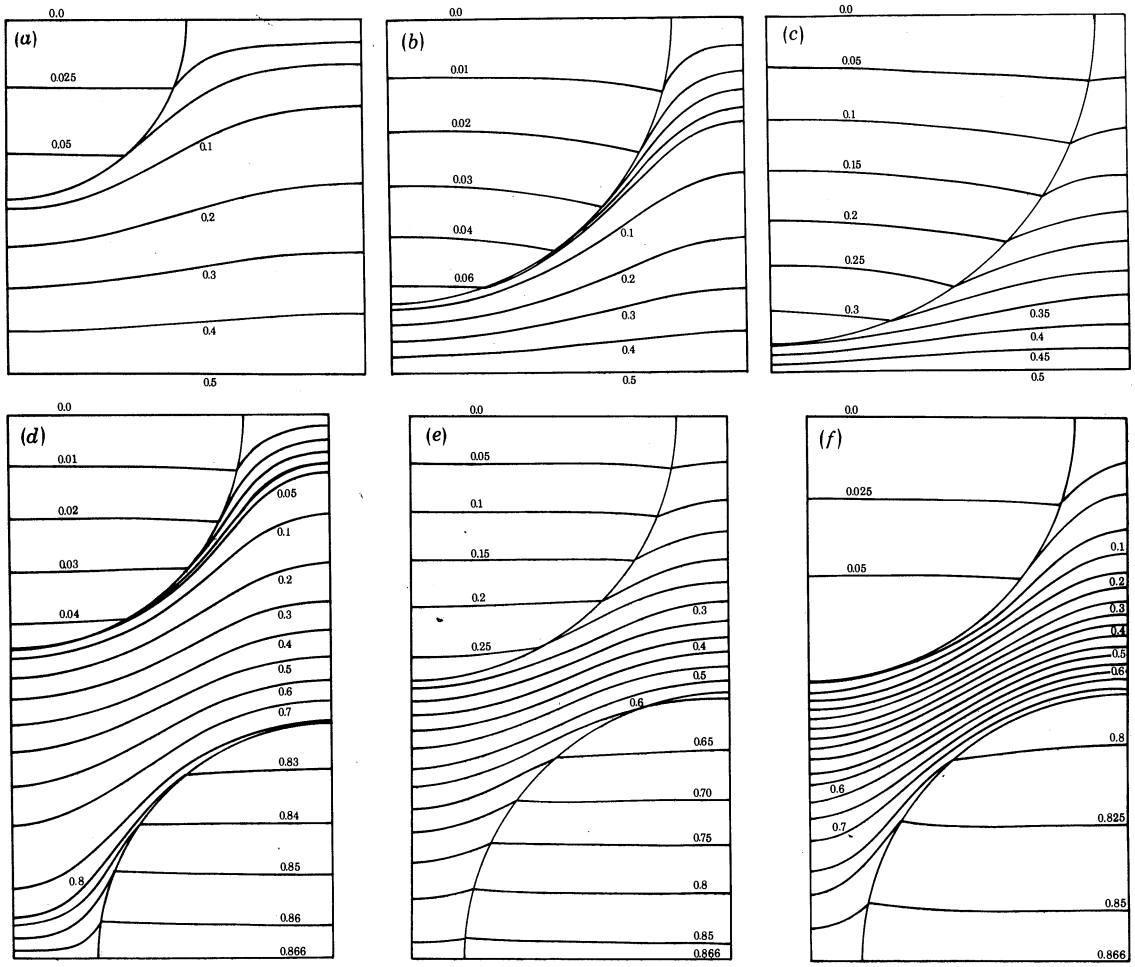



FIGURE 4. Equipotential lines for the square ($a$, $b$, $c$) and hexagonal ($d$, $e$, $f$) arrays, with unit applied field. The respective values of the cylinders conductivity ($\sigma$) and the volume fraction ($f$) are: ($a$), $\sigma = 8.0, f = 0.2$; ($b$), $\sigma = 4.0, f = 0.65$; ($c$), $\sigma = 30.0, f = 0.5$; ($d$), $\sigma = 30.0, f = 0.5$; ($e$), $\sigma = 4.0, f = 0.65$; ($f$), $\sigma = 25.0, f = 0.65$.

For high conductivities, condition (3) implies that the equipotentials cut the cylinder surface at angles close to grazing. This is well exemplified in figure 4 ($c$).

In the case of the hexagonal array, the applied field direction does not coincide with a line of nearest neighbours. This results in a more complex pattern of equipotentials but the phenomenon of field line bunching and the concentration of

Downloaded from rspa.royalsocietypublishing.org



polarization charge is still evident in figures 4(*d*), (*e*), (*f*). For the hexagonal array, the curvatures of the equipotential lines are always small within the cylinders.

For an isolated cylinder placed in a uniform external field, there exists a spatially uniform field everywhere within the cylinder. This fact makes it possible to define a quantity $L$ called the depolarization factor, which is independent of cylinder conductivity (see, for example, Panofsky & Phillips 1964). For a lattice of cylinders, we see from figure 4 that the internal field is no longer homogeneous, so that $L$ can no longer be defined in the standard way. We have investigated generalized definitions of depolarization factor for cylinders in arrays. (For example, we have used a mean internal field in the standard formula for $L$.) Such attempts ended either in a depolarization factor no longer independent of cylinder conductivity $\sigma$, or not related in any simple fashion to the array conductivity $\epsilon$.

We conclude from this work that the interaction effects between cylinders require for their description a complete field expansion of the form (1) and (2). It is not possible to obtain a coherent picture of proximity effects by using formulae based on dipolar field expansions even if the depolarization factor is regarded as a free parameter.

## 7. Reciprocal relations for conductivity

J. B. Keller (1963, 1964) has given an elegant proof of a theorem that relates the conductivity $\epsilon_x$ of an array of cylinders with conductivity $\sigma$ (for an applied field in the $x$-direction) to the conductivity $\epsilon_y$ for the same array but with cylinders of conductivity $1/\sigma$ and the field applied in the $y$-direction.

His result is

$$\epsilon_y = \epsilon_x^{-1}. \tag{18}$$

Keller's proof relies on the properties of conjugate harmonic functions, and holds for a rectangular lattice of identical parallel cylinders, but only for real values of the conductivity $\sigma$. Keller (1964) also states that result (18) also applies to the average conductivity of a statistically homogenous, isotropic random distribution of cylinders.

Our numerical results have indicated that equation (18) holds even for complex values of the conductivity $\sigma$. Furthermore, it is valid for both the square and hexagonal arrays. The latter is a special case of the centred rectangular array rather than the normal rectangular array. The reciprocal property is exactly true for arbitrary orders of solution $N$ of equations (7), whether square or triangular truncation is used. We will give here an algebraic proof of all these properties, for the case of the hexagonal array. The proof requires only slight modification for the square array.

We have already given the system of equations (7) that enable the determination of the unknowns $B_l$ for the problem in which the cylinders have conductivity $\sigma$ and the external field is applied along the direction PI indicated in figure 1. Denote by $B_l'$ the set of unknowns that result when the cylinders have conductivity $\sigma^{-1}$

Downloaded from rspa.royalsocietypublishing.org



and the external field is applied in the orthogonal direction (along the line PG of figure 1). With $S'_l$ to denote the set of lattice sums appropriate to the second problem, the $B'_l$ can be obtained from the set of equations

$$E\delta_{n,1}+(2n-1)!\,\frac{(1+\sigma)}{(1-\sigma)}\,\frac{B'_{2n-1}}{a^{4n-2}}=\sum_{m=1,2,\ldots}\frac{(2n+2m-3)!}{(2l-2)!}\,S'_{2n+2m-2}\,B'_{2m-1}, \tag{19}$$

where $n = 1, 2, \ldots, N$.

The sums $S'_l$ are related to the sums $S_l$ by a rotation through an angle $\phi = \frac{1}{2}\pi$. Hence from the equation A 1.3 of appendix 1,

$$S'_l = S_l e^{-\frac{1}{2}il\pi}, \tag{20}$$

for all values of $l$ greater than 2. For the special case of $l = 2$ the lattice sum $S'_2$ is equal to $S_2$, since it depends simply on the volume density of lattice points (McPhedran & McKenzie 1978) and is independent of the choice of applied field direction.

Consider equation (19) for $n = 1$:

$$E+\frac{(1+\sigma)}{(1-\sigma)}\,\frac{B'_1}{a^2}=S_2\,B'_1+\sum_{m=2,3,\ldots}(2m-1)\,S_{2m}(-1)^m\,B'_{2m-1}. \tag{21}$$

Let us define

$$B''_{2m-1}=(-1)^m B'_{2m-1}. \tag{22}$$

Then

$$[E-2S_2\,B'_1]-\frac{(1+\sigma)}{(1-\sigma)}\,\frac{B''_1}{a^2}=\sum_{m=1,2,\ldots}(2m-1)\,S_{2m}\,B''_{2m-1}. \tag{23}$$

Similarly, equation (19) for $n > 1$ can be written as:

$$-(2n-1)!\,\frac{(1+\sigma)}{(1-\sigma)}\,\frac{B''_{2n-1}}{a^{4n-2}}=\sum_{m=1,2,\ldots}\frac{(2n+2m-3)!}{(2m-2)!}\,S_{2n+2m-2}\,B''_{2m-1}. \tag{24}$$

From equation (11) for the second problem,

$$E-2S_2B'_1=E\epsilon',$$

and so equations (23) and (24) can be combined into a single equation:

$$\epsilon' E\delta_{n,1}-(2n-1)!\,\frac{(1+\sigma)}{(1-\sigma)}\,\frac{B''_{2n-1}}{a^{4n-2}}=\sum_{m=1,2,\ldots}\frac{(2n+2m-3)!}{(2m-2)!}\,S_{2n+2m-2}\,B''_{2m-1}. \tag{25}$$

Comparing equations (7) and (25) and invoking the uniqueness theorem for electrostatic potential, we find

$$B'_{2n-1}=\epsilon'(-1)^n B_{2n-1} \tag{26}$$

for all $n = 1, \ldots, N$. In particular,

$$B'_1=-\epsilon' B_1,$$

and so from (11)

$$\epsilon=1+(2S_2/E)\,(-B'_1/\epsilon'), \tag{27}$$

i.e.

$$\epsilon=1/\epsilon'. \tag{28}$$

Downloaded from rspa.royalsocietypublishing.org



The properties (26) and (28) are true for arbitrary $N$. It can be verified simply that expressions (12) for $\epsilon$ obey (28), and so the reciprocal property is satisfied by the solution of (7), irrespective of the order or fashion of truncation, for both real and complex values of $\sigma$.

A similar relation to (28) does not exist for spheres, since a lattice of perfectly conducting spheres is itself a perfect conductor, whereas the same lattice of zero conductivity spheres has a non-zero conductivity.

## 8. Applications to thin films

It is a commonly observed phenomenon that films deposited onto a substrate by physical vapour deposition (a generic term including the processes of evaporation in vacuum and sputtering by ion bombardment) show a columnar structure (see, for example, Thornton 1977 and Pulker & Jung 1971). The columns commence their growth on nucleation sites on the substrate and grow upward from the substrate, leaving void spaces between. The volume fraction of solid material in the films is usually greater than 0.75 (Ogura 1975; Harris *et al.* 1978). The structure is usually not a regular one but the overall pattern is that of the hexagonal array, each column being roughly hexagonal. In the calculations of this section we will treat the structure as though it were a regular hexagonal array of circular cylinders but in applying the results to real films it must be remembered that the effects of irregularity may be significant, especially for high volume fractions.

Our interest is primarily in the calculation of the refractive index of such structures, including the case where there is a high contrast between the refractive indices of the cylindrical columns and of the matrix. In the cases of high contrast or high volume fractions, a high order $N$ of solution must be used to obtain accurate results.

It will be noted that the results of this section do not explicitly depend upon the scale of the structure relative to the wavelength. Since we are using an electrostatic approach, in order for the results to be valid, the array period must be much smaller than the wavelength. Also, since our calculation of the permittivity assumes the applied electric field to be perpendicular to the cylinder axes, the polarization of the incident electromagnetic wave must fulfil this condition. In the language of optics, columnar structures are uniaxial, with the other principal value of the permittivity or refractive index being calculable from a simple capacitance model (Rayleigh 1892).

First consider the case of a film consisting of columns of copper in air or vacuum. The quantity we shall calculate is the reflectance at normal indicence from the top surface of the array, as a function of wavelength. The surface is cut so that the cylinder axes are normal to it. The columns are taken to be sufficiently long so that all radiation not reflected at the top is absorbed.

In order to calculate the reflectance, the complex refractive index $(n+ik)$ of copper at the wavelength in question (as tabulated in the American Institute of

Downloaded from rspa.royalsocietypublishing.org



Physics Handbook 1972) is used to evaluate the dielectric constant $\sigma$. Equations (7) are solved to find the complex dielectric constant $\epsilon$, whose square root is the effective refractive index $(\tilde{n}+i\tilde{k})$ of the array. The array reflectance for normally incident light is:

$$R = \frac{(\tilde{n}-1)^2+(\tilde{k})^2}{(\tilde{n}+1)^2+(\tilde{k})^2}. \tag{29}$$

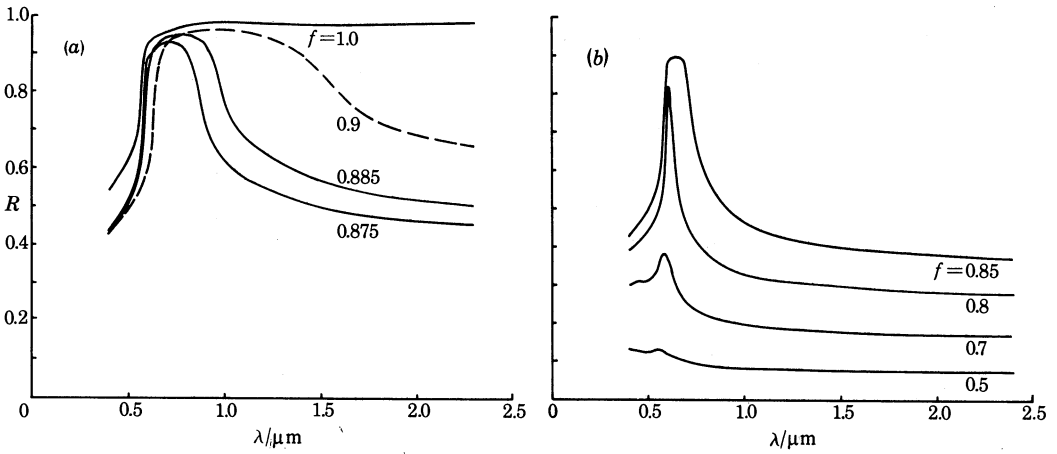



FIGURE 5. (*a*, *b*) The reflectance ($R$) as a function of wavelength ($\lambda$, in μm) is shown for light incident normally on a thick columnar film consisting of circular cylinders of copper in the hexagonal array, with an air matrix. Volume fractions of the cylinders are indicated near the curves.

In figure 5 are shown reflectance curves for seven volume fractions of copper, and also for the bulk material ($f = 1.0$). It can be seen that with decreasing volume fraction, the wavelength region in which the material is a good reflector becomes narrower. The structure becomes a poor reflector at long wavelengths because of the absence of current paths between cylinders. In other words it is a capacitive filter, of the type well known to workers in the far infra-red (Ulrich 1967). As the volume fraction falls below 0.8 the reflectance peak weakens markedly, leaving at volume fractions below 0.5 a structure with the broad absorptance characteristic of a metal black (Harris 1967).

Craig (1979, private communication) has produced copper films with a columnar structure by sputtering copper in argon. The films show reflectance curves resembling those of figure 5.

The second case is the complementary structure consisting of cylindrical voids in a matrix of copper. As a corollary of the reciprocal relations discussed previously an associated relation between reflectances can be obtained. If $R$ is the reflectance of the capacitive structure of effective refractive index $n_c$ at the interface between the structure and air, then the reflectance $R'$ of the complementary structure corresponds to an interface between a medium of index $n_c$ and bulk copper.

Downloaded from rspa.royalsocietypublishing.org



Figure 6 shows the reflectance as a function of wavelength for seven volume fractions of cylindrical voids in copper. The most interesting range of volume fractions is that between 0.8 and 0.9, in which there is a critical dependence of the position of the absorptance edge on volume fraction. At the highest volume fractions the reflectance curve has the characteristics required for a solar selective absorber. We have calculated the solar absorptance of the array with volume fraction 0.9 for the case of sunlight of 'air mass one' to be 0.92. The emittance of the surface of this array at a temperature of 150 °C was calculated to be 0.091. These values constitute good performance figures for a selective surface coating of a solar collector.

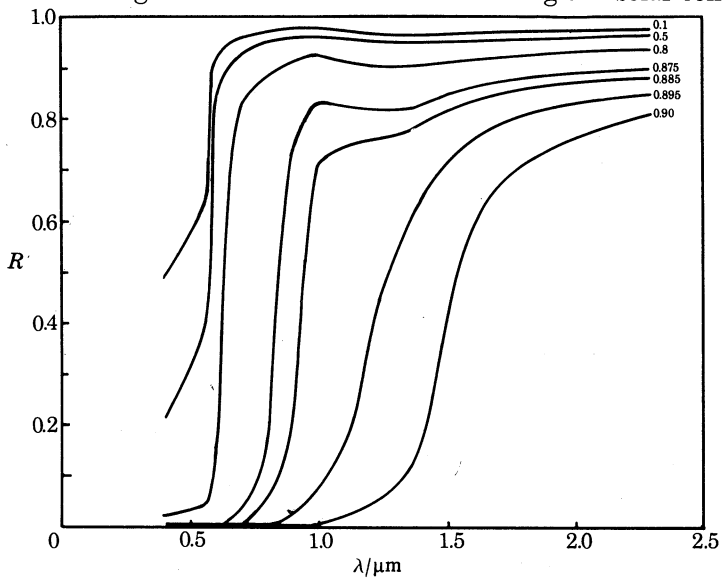



FIGURE 6. $R$ as a function of $\lambda$ (in μm) for light normally incident on a thick film consisting of cylindrical voids in a matrix of copper (hexagonal array), for seven volume fractions of the voids.

The structure of figure 6 has a high reflectance at long wavelengths and corresponds to an inductive filter (Ulrich 1967). It represents the limit as the period tends to zero (with the groove depth remaining large) of the two dimensional gratings studied theoretically by Derrick *et al.* (1979) and, experimentally, by Hutley (1978, private communication). The measurements of Hutley on gratings of period 0.25 μm have shown them to have good solar-selective properties. From figure 6 we can confirm that this solar selectivity persists into the electrostatic region, provided that the bosses on the grating surface are densely packed.

## 9. Conclusions

Our previous work on lattices of spheres and the work reported here show the power and versatility of our extension of Lord Rayleigh's method for calculating transport properties. It has already been applied to a variety of situations, and has

Downloaded from rspa.royalsocietypublishing.org



been in far better accord with experiment than have calculations based on the (incomplete) dipolar model. We would advocate its widespread use.

The authors thank Professor H. Messel and the Science Foundation for Physics for their support. This work was undertaken while one of the authors (R. C. McPhedran) was the recipient of an Australian Research Grants Committee Fellowship.

The referee is thanked for his comments, which have led to an improved version of the paper.

## Appendix 1. Evaluation of the sums for the hexagonal array

Let us consider the properties of the sums

$$S_l = \sum_{(x,y)} 1/(x+\mathrm{i}y)^l, \tag{A 1.1}$$

where $l = 2, 3, \ldots$ and where the points $(x, y)$ trace out all cylinder centres in a two dimensional array. Introducing polar coordinates $(r, \theta)$, (A 1.1) becomes

$$S_l = \sum_{(r,\theta)} (1/r^l)\,\mathrm{e}^{-\mathrm{i}l\theta}. \tag{A 1.2}$$

Suppose that the array is unaltered by an angular rotation through an angle $\phi$. Then:

$$S_l = \sum_{(r,\theta)} (1/r^l)\,\mathrm{e}^{-\mathrm{i}l(\theta+\phi)} = S_l\,\mathrm{e}^{-\mathrm{i}l\phi} \tag{A 1.3}$$

for all integral values of $l > 2$. In order that $S_l$ be non-zero we must have

$$l = 2\pi m/\phi \tag{A 1.4}$$

for some integer $m$.

Turning now to the particular case of the hexagonal array (figure 1), $\phi = \frac{1}{3}\pi$ and so the only non-zero sums have an order $l$ which is a multiple of 6 (apart from $S_2$, which was considered in the body of the paper). The sums $S_6$, $S_{12}$, ... have been evaluated by direct summation. We give their values here, accurate to five decimal places:

$$\left.\begin{aligned} S_2 &= \tfrac{2\pi}{\sqrt{3}} = 3.62760, \\ S_6 &= -5.86303, \\ S_{12} &= 6.00964, \\ S_{18} &= -5.99972, \\ S_{24} &= 6.00001, \\ S_{30} &= S_{42} = S_{54} = \ldots = -6.00000, \\ S_{36} &= S_{48} = S_{60} = \ldots = 6.00000. \end{aligned}\right\} \tag{A 1.5}$$

The hexagonal array can be viewed as the union of two identical sub-arrays, one of which is displaced through the vector $(\frac{\sqrt{3}}{2}, \frac{1}{2})$ in the $x, y$ plane relative to the other.

Downloaded from rspa.royalsocietypublishing.org



The following approximate formulae give the contributions to the sums from each sub-array:

$$S^1_{6l} = 2\left[\left((-1)^l+\frac{1}{3^{3l}}\right)\left(\zeta(6l)+\frac{2}{4^{3l}}\right)+\frac{2\cos(6l\theta_{12})}{7^{3l}}\right], \tag{A 1.6}$$

and

$$S^2_{6l} = 4\left[(-1)^l+\frac{1}{3^{3l}}+\frac{(-1)^l}{3^{6l}}+\frac{2}{7^{3l}}\cos(6l\psi_{1,0})+\frac{2}{13^{3l}}\cos(6l\psi_{1,2})\right]. \tag{A 1.7}$$

Here $\zeta(6l)$ denotes the Riemann zeta function of argument $6l$, and

$$\theta_{12} = \arccos(\sqrt{\tfrac{3}{7}}), \tag{A 1.8}$$

$$\left.\begin{aligned}\psi_{1,0} &= \arccos(\tfrac{3}{2}\sqrt{\tfrac{3}{7}}),\\ \psi_{1,2} &= \arccos(\tfrac{3}{2}\sqrt{\tfrac{3}{13}}).\end{aligned}\right\} \tag{A 1.9}$$

These two formulae when added give $S_{6l}$ correct to five decimal places for $l > 1$.

## Appendix 2. Evaluation of sums for the square array

The square array is invariant under rotations through integral multiples of $\frac{1}{2}\pi$, so that (apart from $S_2$) all non-zero sums have $l$ divisible by four. By direct summation, we have obtained the following values (correct to five decimal places):

$$S_2 = \pi = 3.14159, \tag{A 2.1}$$
$$S_4 = 3.15121 = \pi^4A^4/60, \tag{A 2.2}$$
$$S_8 = 4.25577 = \pi^8A^8/(70\times120), \tag{A 2.3}$$
$$S_{12} = 3.93885 = \pi^{12}A^{12}/(110\times120\times130), \tag{A 2.4}$$
$$S_{16} = 4.015695, \tag{A 2.5}$$
$$S_{20} = 3.99610, \tag{A 2.6}$$
$$S_{24} = 4.00098, \tag{A 2.7}$$
$$S_{28} = 3.99976, \tag{A 2.8}$$
$$S_{32} = 4.00006, \tag{A 2.9}$$
$$S_{36} = S_{40} = S_{44} = \ldots = 4.000\,00. \tag{A 2.10}$$

The analytic expressions for $S_4$ and $S_8$ were obtained by Lord Rayleigh (1892), with

$$A = 1.18034060. \tag{A 2.11}$$

The sums $S_{4l}$ for $l > 2$ may be obtained (correct to five decimal places) from the formula:

$$S_{4l} \simeq 4\left[\zeta(4l)\left(1+\frac{(-1)^l}{2^{2l}}\right)+\frac{2\cos(4l\theta_{2,1})}{5^{2l}}\right], \tag{A 2.12}$$

where

$$\theta_{2,1} = \arccos(2/\sqrt{5}). \tag{A 2.13}$$

Downloaded from rspa.royalsocietypublishing.org



## References

*American Institute of Physics Handbook* 1972 New York: McGraw-Hill.
Batchelor, G. K. 1974 *A. Rev. Fluid Mech.* **6**, 227–255.
Derrick, G. H., McPhedran, R. C., Maystre, D. & Nevière, M. 1979 *Appl. Phys.* **18**, 39–52.
Desorbo, W. & Kline, H. E. 1970 *J. appl. Phys.* **41**, 2099–2105.
Doyle, W. T. 1978 *J. appl. Phys.* **49**, 795–797.
Galeener, F. L. 1971 *Phys. Rev.* **27**, 321–423.
Harris, L. 1967 *The optical properties of metal blacks and carbon blacks*, Eppley Foundation for Research Monograph Series, no. 1. Newport, R. I.
Harris, M., Macleod, H. A., Ogura, S., Pelletier, E. & Vidal, B. 1978 *Fourth International Thin Films Conference*, Loughborough, England.
*International tables for X-ray crystallography* 1962 Birmingham: Kynoch Press.
Keller, J. B. 1963 *J. appl. Phys.* **34**, 991–993.
Keller, J. B. 1964 *J. math. Phys.* **5**, 548–549.
Kharadly, M. M. Z. & Jackson, W. 1952 *Proc. Instn elect. Engrs* **100**, 199–212.
McKenzie, D. R., McPhedran, R. C. & Derrick, G. H. 1978 *Proc. R. Soc. Lond.* A **362**, 211–322.
McPhedran, R. C. & McKenzie, D. R. 1978 *Proc. R. Soc. Lond.* A **359**, 45–63.
Ninham, B. W. & Sammut, R. A. 1976 *J. theor. Biol.* **56**, 125–149.
Netterfield, E. P. 1976 *Appl. Opt.* **15**, 1969–1973.
O'Brien, R. W. 1977 Ph.D. thesis, University of Cambridge.
Ogura, S. 1975 Ph.D. thesis, Newcastle-upon-Tyne Polytechnic.
Panofsky, W. K. H. & Phillips, M. 1962 *Classical electricity and magnetism*, 2nd edn. Reading, Mass: Addison-Wesley.
Perrins, W. T., McPhedran, R. C. & McKenzie, D. R. 1979 *Thin Solid Films* (to appear).
Pulker, H. K. & Jung, E. 1971 *Thin Solid Films* **9**, 57–66.
Rayleigh, Lord 1892 *Phil. Mag.* **34**, 481–502.
Runge, I. 1925 *Z. tech. Phys.* **6**, 61–68.
Thornton, J. A. *A. Rev. mater. Sci.* **7**, 239–260.
Ulrich, R. 1967 *Infrared Phys.* **7**, 37–55.
Visscher, W. M. & Bosterli, M. 1972 *Nature, Lond.* **239**, 504–507.

PROCEEDINGS OF THE ROYAL SOCIETY A MATHEMATICAL, PHYSICAL & ENGINEERING SCIENCES
PROCEEDINGS OF THE ROYAL SOCIETY A MATHEMATICAL, PHYSICAL & ENGINEERING SCIENCES